Superpixel-Level Weighted Label Propagation for Hyperspectral Image Classification

Sen Jia, *Senior Member, IEEE*, Xianglong Deng, Meng Xu, *Member, IEEE*, Jun Zhou, *Senior Member, IEEE*, and Xiuping Jia, *Senior Member, IEEE*

***Abstract*— As a typical graph-based semisupervised learning technique, the label propagation (LP) approach has gained much attention in recent years. The key to LP algorithms is the propagation capability and efficiency of the similarity matrix, which describes the similarity between two data points. Concerning hyperspectral image which often contains hundreds of thousands of pixels, the corresponding similarity matrix is particularly huge and thus the LP procedure is intractable. Fortunately, superpixel, which can effectively characterize the spatial semantic information of surface objects, provides a reasonable way to solve this problem. In this article, we propose an elaborate superpixel-based weighted LP approach, abbreviated as SuWLP, for hyperspectral image classification. First, the hyperspectral image is oversegmented by the entropy rate segmentation (ERS) method, and the internal consistency of each superpixel can be achieved. Second, a new similarity measure is designed to estimate the similarity between two superpixels, and a superpixel-based similarity matrix can be thus established. Third, after the training samples have been expanded based on the superpixel distribution, a weighted LP technique is designed to propagate the sample label at the superpixel level without any parameter tuning. Finally, the label of each superpixel maps back to the contained pixels. We compared our proposed SuWLP method with several state-of-the-art ones, and experimental results on three real hyperspectral data sets certify the effectiveness and efficiency of the superpixel-level LP strategy.**

***Index Terms*— Hyperspectral image, label propagation (LP), superpixel segmentation.**

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I. Introduction

In HYPERSPECTRAL image literature, many supervised classifiers have been introduced and investigated to tackle the hyperspectral image classification problem [1], [2]. For traditional supervised learning strategies, abundant labeled samples are generally required to adequately train the classifier. However, the cost of attaining labeled samples is very expensive and time-consuming actually [3], and thus the unbalanced and limited labeled samples are always insufficient to estimate the parameters of the classifier well for hyperspectral image. Fortunately, since unlabeled data are easy to acquire compared to labeled data, semisupervised learning can effectively exploit both labeled and unlabeled samples and the small size sample problem can be well defined [4].

Technically, semisupervised learning takes the advantages of both supervised learning and unsupervised learning, in which labeled and unlabeled data are integrated together. Clustering hypothesis, manifold hypothesis, and global–local consistency hypothesis are three commonly used basic hypotheses for semisupervised learning [5]. Essentially, these three assumptions are consistent with each other, that is, the more similar the data, the more likely they have the same label. According to the classification strategy of semisupervised learning algorithms, they can be roughly categorized into the following five types.

1) *Transductive Support Vector Machine (TSVM) [6]:* TSVM considers both labeled and unlabeled samples to minimize the generalization error and determine the optimal classification boundary.
2) *Generative Model (GM) [7]–[11]:* Through learning the density function from labeled samples, GM classifies all unlabeled samples based on the Bayesian probability theorem.
3) *CoTraining [12]–[14]:* Two classifiers are pretrained by the labeled data, and the classification results of testing samples generated by one classifier are subsequently used as the newly added labeled data for the next iterative training process of another classifier.
4) *Self-Training [15], [16]:* Self-training methods first utilize labeled samples to train the initial classifier as usual, and the most reliable labeled ones identified by the initial classifier are put into the existing training sample set. The new training sample set is then used for the next classification process. This procedure is repeated until the cutoff condition is met.

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5) *Graph-Based Semisupervised Learning [17]–[19]:* A graph is first formulated to represent the data similarity, in which the vertices denote samples, whereas the edges between points characterize the similarity of samples, and semisupervised learning is then applied on the graph. These algorithms have demonstrated good performance in many areas (including community discovery, digital recognition, image classification, and so on), and research on semisupervised learning has become active in recent years.

Considering graph-based semisupervised learning methods, Zhu [20] proposed semisupervised learning with a graph, which is based on an assumption that nodes connected by edges with large similarity are likely to have the same label. Wang and Zhang [21] proposed a neighborhood-based model to propagate labels from labeled data to unlabeled ones through linear neighborhood propagation (LNP). Further, Tian and Kuang [22] presented a novel method called global linear neighborhoods label propagation (GLNP), which learns a nonnegative low-rank graph to capture global linear neighborhoods. Alternatively, it can be observed that the essence of the graph-based approach is label propagation (LP). Based on the manifold hypothesis and the geometry structure among samples, the class label propagates from labeled samples to unlabeled ones according to the adjacency relationship on the graph. Specifically, in order to reflect the clustering relationship between labeled and unlabeled data, LP was proposed by Yang *et al.* [23], which takes nonnegative sparse representation into account. Furthermore, a dynamic version of the LP algorithm [dynamic label propagation (DLP)] was proposed by Wang and Tsotsos [24], and DLP has a higher convergence speed than LP.

Obviously, in graph-based semisupervised learning algorithms, the most important issue is the propagation capability and efficiency of the similarity matrix and the construction of the graph. Traditionally, the construction of a graph mainly contains three steps. First, a suitable distance measure, such as Euclidean distance and Chebyshev distance, is chosen to calculate the similarity between samples. Second, a feasible scheme is selected to construct the connection diagram, which can be divided into a dense graph and a sparse graph while the latter is more common. For a sparse graph, there are two popular computational methods: k-nearest neighbors and ε-nearest neighbors [25]. Third, the kernel function is used to assign weights to the connected edges in the graph, and the edge weight reflects the similarity between points. However, concerning hyperspectral image classification, which intends to designate a unique class label to each spatial pixel [26], [27], the time and spatial complexity of constructed pixel-based similarity matrix is usually infeasible. For example, the popular Salinas hyperspectral image data set contains more than 100 000 pixels (that is, the spatial dimension is 512×217) and the storage for similarity matrix exceeds 90 GB, leading to the intractability of graph-based semisupervised learning methods.

Fortunately, the distribution of surface materials usually exhibits regularly in space, and thus the samples in the local area are likely to have the same label. Therefore, the concept of superpixel can be introduced to handle the problem, which reflects the phenomenon that pixels in the local area have similar attributes, including color, brightness, and texture [28]. In fact, the superpixel idea has been used for preprocessing and postprocessing of hyperspectral image processing [29], [30]. In terms of preprocessing, superpixel segmentation is performed and then feature extraction is carried out on the superpixel level [31], [32], which can greatly reduce the computational burden. Alternatively, superpixel-based postprocessing techniques mostly use superpixels to regularize the results obtained by other classifiers [33], [34]. Likewise, assuming that each superpixel contains only ten pixels, the space complexity of a superpixel-based similarity matrix is only one-hundredth of the pixel-based one, and thus the space storage and computational complexity can be significantly decreased. Hence, it is desirable to incorporate superpixel segmentation into graph-based semisupervised learning to reduce the scale of the graph and the spatial and computational demanding can be well resolved.

In this article, an elaborate superpixel-based weighted LP approach, abbreviated as SuWLP, has been proposed for hyperspectral image classification. First, a superpixel segmentation method called entropy rate segmentation (ERS) is utilized to segment the hyperspectral image to attain an oversegmented superpixel map, and the homogeneity within each superpixel can be well retained. Second, a new similarity measure is designed to estimate the similarity between two superpixels with an irregular shape, and a superpixel-based similarity matrix can thus be established. Third, after the training samples have been expanded based on the superpixel distribution, a weighted LP technique is designed to propagate the sample label at the superpixel level without any parameter tuning. Finally, the label of each superpixel maps back to the contained pixels. To make the proposed SuWLP procedure more easily to be understood, a schematic of SuWLP is displayed in Fig. 1. Meanwhile, different from the traditional graph-based semisupervised learning algorithms, our SuWLP approach has the following three main contributions:

1) First of all, we propose to propagate the sample labels in a superpixel-level graph rather than pixel-level one, which not only significantly increases the efficiency of the LP procedure (both time and space), but also takes full advantage of the spatial–structural semantic information of surface objects. Due to the homogeneity within each superpixel and powerful learning capability of the proposed weighted LP method, the classification performance of our SuWLP is guaranteed.
2) The hyperspectral image has been oversegmented by the ERS method, hence the pixels within each superpixel most likely belong to the same class, and the superpixel homogeneity can be well retained. Meanwhile, since the spatial distribution of every superpixel is irregular, a new distance measure is introduced here to characterize the similarity of superpixels in the neighborhood (instead of the widely used Euclidean and Gaussian distance), and the representative ability of the constructed propagation map can thus be enhanced.
3) Rather than manually specifying the parameters contained in the LP algorithm, we design a weighted LP technique

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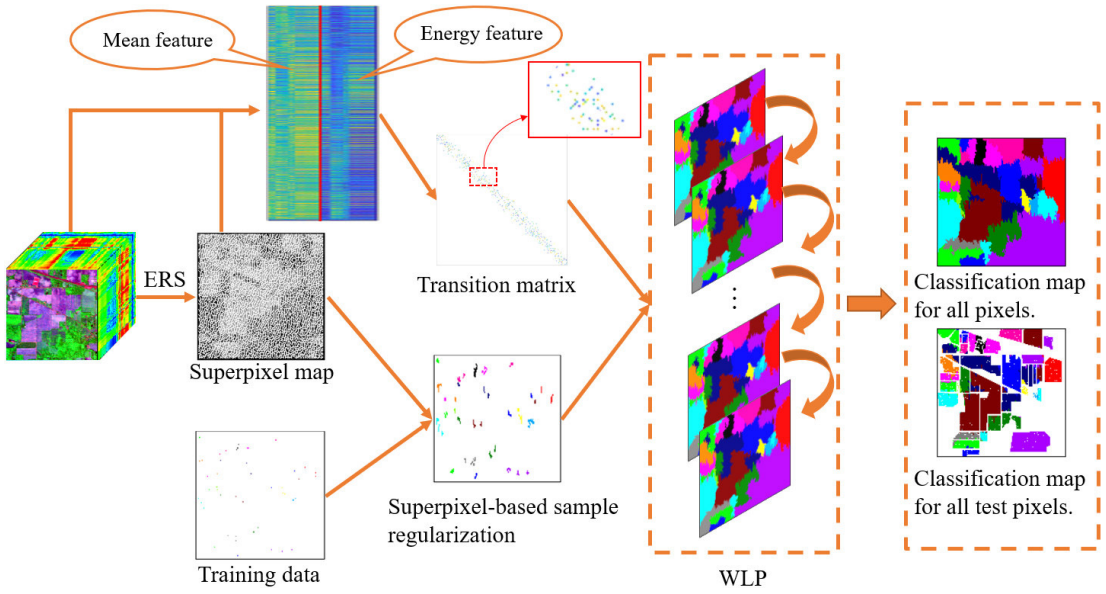


Fig. 1. Proposed SuWLP procedure for hyperspectral image classification.

to automatically determine the parameters by investigating the statistical properties of extracted superpixels. Likewise, the number of extracted superpixels can also be computed according to the intrinsic property of the hyperspectral image (i.e., the spatial coverage and resolution). Hence, the robustness of the proposed SuWLP approach can be ensured, and it can be directly applied to hyperspectral images with various spectral and spatial resolutions without any parameter tuning process.

Several state-of-the-art methods are considered in comparison, and experimental results on three real hyperspectral image data sets certify the effectiveness and efficiency of the superpixel-level LP strategy.

The remainder of this article is organized as follows. Section II briefly reviews the LP, DLP and superpixel segmentation. Section III presents the proposed SuWLP approach in detail. Section IV lists the information of three real hyperspectral image data sets and describes the experimental design. The experimental results are provided in Section V. Finally, conclusions are given in Section VI.

II. Related Works

In this section, we will briefly review some of the works that are related to this article.

A. Label Propagation (LP)

Given a data set $\mathbf{X} = \{\mathbf{x}_i, i = 1, \ldots, N\}$, the corresponding finite weighted graph can be constructed as $G = (V, E, \mathbf{W})$, where V denotes the vertex set, $E \subseteq V \times V$ is a set of edges, and $\mathbf{W} \in \mathbb{R}^{N\times N}$ is a nonnegative symmetric similarity matrix. In general, the similarity between any two vertices $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined as

$$\mathbf{W}(i, j) = h\left(\frac{\|\mathbf{x}_i, \mathbf{x}_j\|_2^2}{\theta^2}\right) \tag{1}$$

where $h(x)$ is an exponential decay function and usually in the form of $\exp(-x)$, and $\|\cdot\|_2$ is the l_2-norm distance function. Parameter θ is the hyper-parameter and equal to the mean distance of K-nearest neighborhoods. More details are given in [35].

Furthermore, from the obtained similarity matrix, $\mathbf{W}$, the transition matrix, $\mathbf{P}$, on vertices, V, is calculated by

$$\mathbf{P}(i, j) = \frac{\mathbf{W}(i, j)}{\sum_{k\in V} \mathbf{W}(i, k)}. \tag{2}$$

Clearly, after the normalization operation, $\mathbf{P}$ is asymmetric and $\sum_{j\in V} \mathbf{P}(i, j) = 1$.

Since LP is a semisupervised leaning method, the label of training samples propagates to the unlabeled ones through the constructed transition matrix, $\mathbf{P}$. Specifically, the data set $\mathbf{X}$ is divided into two nonoverlapped parts, in which $\mathbf{X}^{(l)} = \{\mathbf{x}_i, i = 1, \ldots, N_l\}$ represents the labeled samples and $\mathbf{X}^{(u)} = \{\mathbf{x}_i, i = N_l + 1, \ldots, N\}$ denotes the unlabeled data (the number of unlabeled samples N_u equals to $N - N_l$). For multiclass LP, let $\mathbf{Y} = [\mathbf{Y}^{(l)}; \mathbf{Y}^{(u)}] \in \mathbb{R}^{N\times C}$ stand for the label probability matrix, where C is the number of classes. Likewise, $\mathbf{Y}^{(l)} \in \mathbb{R}^{N_l\times C}$ and $\mathbf{Y}^{(u)} \in \mathbb{R}^{N_u\times C}$, respectively, describes the label probability of labeled and unlabeled samples. Obviously, since the labels in $\mathbf{X}^{(l)}$ are known in advance, the value of $\mathbf{Y}^{(l)}$ is set as

$$\mathbf{Y}^{(l)}_{(i,c)} = \begin{cases} 1, & \text{if } \mathbf{x}_i \text{ is labeled as class } c,\ \mathbf{x}_i \in \mathbf{X}^{(l)} \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Given the definition of related symbols, the main procedure of LP is summarized as follows.

1) Construct the transition matrix $\mathbf{P}$ by (2).
2) Initialize label probability matrix $\mathbf{Y}_0 = [\mathbf{Y}_0^{(l)}; 0]$, where $\mathbf{Y}_0^{(l)}$ is set by (3).
3) Perform the following steps ($t = 1, \ldots, \mathrm{T}$) until $\mathbf{Y}_\mathrm{T}$ converges to a stable condition.
   a) $\mathbf{Y}_t = \mathbf{P} * \mathbf{Y}_{t-1}$.
   b) $\mathbf{Y}_t^{(l)} = \mathbf{Y}_0^{(l)}$.
4) Output $\mathbf{Y}_\mathrm{T}$.

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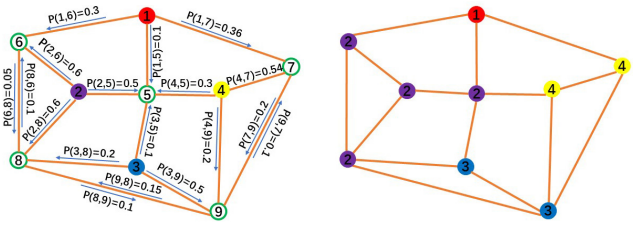


Fig. 2. Procedure of LP in one iteration. (a) Constructed graph based on transition matrix $\mathbf{P}$ before propagation. The solid node with different color corresponds to labeled samples and the number in node is the class label, while hollow nodes stand for the unlabeled samples and the number in node is just a sequential numbering. $\mathbf{P}(i, j)$ on the directed edges indicate that the label of $\mathbf{x}_i$ propagates to $\mathbf{x}_j$ with a probability of $\mathbf{P}(i, j)$. (b) Graph after propagation. The number in the node is the predicted class. For instance, according to the transition matrix $\mathbf{P}$ in Fig. 2(a), the labeled samples "1"–"4," respectively, propagate their labels to unlabeled sample "5" with probabilities 0.1, 0.5, 0.1, and 0.3, and the label of sample "5" is labeled as class "2" after propagation because of the transition probability $\mathbf{P}(2, 5)$ is the largest. The other unlabeled samples are predicted in the same way.

Here, the stable condition means that the sample labels are kept unchanged in the propagation procedure, which is judged by the change rate of sample labels in two successive label probability matrix $\mathbf{Y}_t$ and $\mathbf{Y}_{t-1}$. After obtaining the final label probability matrix $\mathbf{Y}_{\mathrm{T}}$, the label of the ith $(i = 1, \ldots, N_u)$ unlabeled sample is predicted to the class that has the maximal value in the $(N_l + i)$th row of $\mathbf{Y}_{\mathrm{T}}$. In order to make the description of LP more clear, the procedure of LP in a single iteration is displayed in Fig. 2. Based on the constructed graph weighted by transition matrix $\mathbf{P}$, the labeled samples propagate to the unlabeled ones by the LP algorithm.

B. Dynamic Label Propagation (DLP)

DLP is a dynamic version of LP that has a faster convergence rate than LP and demonstrates better performance on several computer vision problems [24]. Different from the similarity matrix $\mathbf{W}$ [defined in (1)] that is a symmetric matrix used in LP, the similarity matrix $\mathcal{W}$ utilized in DLP is defined in a more reasonable way. Based on the assumption that similarities in the local region are more reliable than faraway ones, a KNN graph $G_{\mathrm{KNN}} = (V, \mathcal{E}, \mathcal{W})$ (the edge set in G_{KNN} is exactly the same as G) is first constructed to weight edges nearby. In other words, the weight value between neighboring points in the similarity matrix $\mathbf{W}$ are reserved, whereas those between nonneighboring points are set to zero. Then, the similarity matrix $\mathcal{W}$ in the G_{KNN} graph can be defined as

$$\mathcal{W}(i, j) = \begin{cases} \mathbf{W}(i, j), & \text{if } \mathbf{x}_j \in \mathrm{KNN}(\mathbf{x}_i) \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Correspondingly, the normalized transition matrix $\mathbf{P}$ turns into a KNN-based matrix $\mathcal{P}$

$$\mathcal{P} = \frac{\mathcal{W}(i, j)}{\sum_{\mathbf{x}_k \in \mathrm{KNN}(\mathbf{x}_i)} \mathcal{W}(i, k)}. \tag{5}$$

However, the label information from labeled points can only propagate to their neighbors in the local area, hence those nonlocal points require another way to obtain label information. DLP provides an effective method to fuse the information among data feature and data labels, which contains two steps (kernel fusion and kernel diffusion).

According to the assumption that two samples with highly correlated label vectors tend to have high similarity in the input data space, it is reasonable and necessary to investigate the transition matrix and label probability matrix together. Concretely, the fusion operation of the transition matrix $\mathbf{P}_t$ and the label probability matrix $\mathbf{Y}_t$ is

$$\mathbf{F}_t = \mathbf{P}_t + \lambda \mathbf{Y}_t \mathbf{Y}_t^T \tag{6}$$

where $\mathbf{Y}_t^T$ denotes the transpose of $\mathbf{Y}_t$, and λ is a weight coefficient. However, the effect of fusion operation can be decreased along with the iteration, resulting in the nonpropagation of label information. To deal with this problem, a diffusion operator is introduced

$$\mathbf{P}_{t+1} = \mathcal{P} \mathbf{F}_t \mathcal{P}^T + \beta \mathbf{I} \tag{7}$$

where $\mathbf{I} \in \mathbb{R}^{N \times N}$ is an identity matrix, and β is a parameter to weight matrix $\mathbf{I}$.

Likewise, the algorithm of DLP is summarized as follows.

1) Construct transition matrix $\mathbf{P}_0$ by (2).
2) Initialize label probability matrix $\mathbf{Y}_0 = [\mathbf{Y}_0^{(l)}; 0]$, where $\mathbf{Y}_0^{(l)}$ is set by (3).
3) Calculate the KNN-based transition matrix $\mathcal{P}$ by (5).
4) Perform the following steps $(t = 1, \ldots, \mathrm{T})$ until $\mathbf{Y}_{\mathrm{T}}$ converges to a stable condition.
   a) $\mathbf{Y}_t = \mathbf{P}_{t-1} * \mathbf{Y}_{t-1}$.
   b) $\mathbf{Y}_t^{(l)} = \mathbf{Y}_0^{(l)}$.
   c) $\mathbf{P}_t = \mathcal{P}(\mathbf{P}_{t-1} + \lambda \mathbf{Y}_{t-1} \mathbf{Y}_{t-1}^T) \mathcal{P}^T + \beta \mathbf{I}$.
5) Output $\mathbf{Y}_{\mathrm{T}}$.

Here, $(\cdot)^T$ stands for the transpose operation. After the label probability matrix $\mathbf{Y}_{\mathrm{T}}$ is achieved, the label of unlabeled samples is determined the same as LP.

C. Superpixel Extraction for Hyperspectral Image

Superpixel technology has been extensively researched in the computer vision field, in which ERS [36] is one of the widely used superpixel extraction algorithms. Just like the graph construction procedure of LP, an undirected graph $G = (V, E)$ is first built for the ERS method, where V is the vertex set corresponding to pixels in the hyperspectral image, and E is the edge set. Similarly, the edge weights measure the similarity of neighboring pixels, which can be directly computed by (1). In the following, the superpixel segmentation task can be considered as the problem of how to choose a suitable subset $S \subseteq E$ that could actively divide the graph G into K connected subgraphs. More concretely, ERS transfers the segmentation task into optimizing the following criterion:

$$\arg\max_{S} \quad \mathcal{H}(S) + \alpha \mathcal{B}(S), \quad \text{s.t} \quad S \subseteq E \tag{8}$$

where the entropy rate term $\mathcal{H}(S)$ aims to achieve homogeneous clustering, and the role of balancing term $\mathcal{B}(S)$ makes the clusters have a similar size. The regularization parameter α is a nonnegative coefficient that weights the balancing term.

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III. SuWLP Approach

In this section, the proposed SuWLP approach is described in detail. First, the superpixel-oriented feature extraction procedure is presented. Subsequently, the training samples are expanded into the superpixel scale. Third, the transition matrix is constructed from the obtained superpixel map and superpixel-based features, where a new distance measure is introduced. Finally, a weighted LP algorithm is designed on a superpixel level, and the superpixel propagation result is then mapped back to the contained pixels.

A. Superpixel-Oriented Feature Extraction

Some symbols for hyperspectral image are given for convenience. Let $\mathbf{R} \in \mathbb{R}^{X\times Y\times B}$ express the raw hyperspectral image (X and Y are two spatial dimensions, and B is the number of spectral bands). As described in Section II-C, a 2-D superpixel map $\mathbf{S} \in \mathbb{R}^{X\times Y}$ can be generated by applying the ERS method on the raw hyperspectral image $\mathbf{R}$. In our superpixel-level LP method, it is desirable and necessary to ensure the homogeneity within every superpixel, thus the superpixel map $\mathbf{S}$ should be oversegmented. Fortunately, Jia *et* al. [34] provide a feasible way to adaptively determine the number of extracted superpixels K

$$K = \text{floor}\left(\frac{X\times Y}{\text{floor}(50\times 0.7^{\sqrt{\text{Res}}})}\right) \tag{9}$$

where Res is the spatial resolution of hyperspectral image, which is the basic property of hyperspectral image. Meanwhile, the floor$(\cdot)$ operation rounds the value down to the nearest integer. Consequently, as the number of extracted superpixels K is calculated by (9), the superpixel map $\mathbf{S}$ in oversegmentation scene can be achieved by the ERS method, and the kth superpixel $\mathbf{S}_k, k = 1, \ldots, K$, is

$$\mathbf{S}_k = \{\mathbf{S}_{(i,j)} | \mathbf{S}_{(i,j)} = k\}, \quad k = 1, \ldots, K. \tag{10}$$

Clearly, the superpixel map $\mathbf{S}$ can also be described as follows:

$$\mathbf{S} = \bigcup_{k=1}^{K} \mathbf{S}_k. \tag{11}$$

After the superpixel map $\mathbf{S}$ has been obtained, two statistical descriptors, feature mean and energy, are adopted to characterize the superpixel property and thus accelerate the following LP procedure (it is worth pointing out that higher-order statistics, such as Skewness and Kurtosis which, respectively, characterizes the degree of asymmetry and non-Gaussianity of sample distribution, can also be incorporated to represent superpixel). Specifically, the bth element $\mathbf{M}_k^b, b = 1, \ldots, B$, of the feature mean vector $\mathbf{M}_k \in \mathbb{R}^B$ corresponding to the kth superpixel $\mathbf{S}_k, k = 1, \ldots, K$ is computed by

$$\mathbf{M}_k^b = \frac{1}{m_k}\sum_{i,j} \mathbf{R}_{(i,j,b)}, \quad \mathbf{S}_{(i,j)} = k \qquad k = 1, \ldots, K, \quad b = 1, \ldots, B \tag{12}$$

where m_k is the number of pixels inside the kth superpixel $\mathbf{S}_k$. Obviously, the feature mean vector $\mathbf{M}_k \in \mathbb{R}^B$ is the concatenation of $\mathbf{M}_k^b, b = 1, \ldots, B$, that is,

$$\mathbf{M}_k = \left[\mathbf{M}_k^1, \mathbf{M}_k^2, \ldots, \mathbf{M}_k^B\right]^T. \tag{13}$$

In a like manner, the feature energy vector $\mathbf{E}_k \in \mathbb{R}^B$ of the kth superpixel $\mathbf{S}_k, k = 1, \ldots, K$ can be acquired by

$$\mathbf{E}_k = \left[\mathbf{E}_k^1, \mathbf{E}_k^2, \ldots, \mathbf{E}_k^B\right]^T \tag{14}$$

where the bth element $\mathbf{E}_k^b, b = 1, \ldots, B$ can be attained in the same way, that is,

$$\mathbf{E}_k^b = \frac{1}{m_k}\sum_{i,j} \mathbf{R}_{(i,j,b)}^2, \quad \mathbf{S}_{(i,j)} = k \qquad k = 1, \ldots, K, \quad b = 1, \ldots, B. \tag{15}$$

As the feature mean and energy vectors, respectively, $\mathbf{M}_k$ and $\mathbf{E}_k$, have been achieved for the kth superpixel $\mathbf{S}_k, k = 1, \ldots, K$, and they are combined together to characterize the kth superpixel $\mathbf{S}_k$, that is

$$\mathbf{G}_k = [\mathbf{M}_k; \mathbf{E}_k], \quad k = 1, \ldots, K. \tag{16}$$

At last, the feature matrix $\mathbf{G} \in \mathbb{R}^{K\times 2B}$ for the whole superpixel map $\mathbf{S}$ is

$$\mathbf{G} = [\mathbf{G}_1, \mathbf{G}_2, \ldots, \mathbf{G}_K]^T. \tag{17}$$

B. Superpixel-Level Sample Regularization

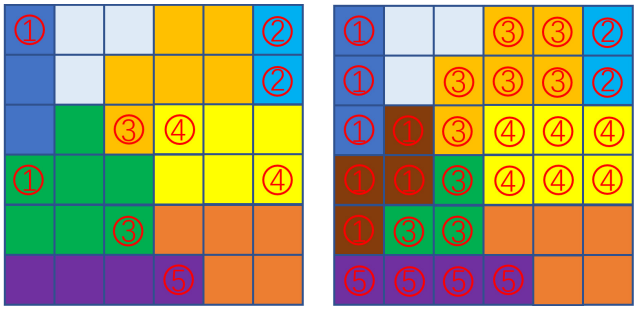
Since the LP method is carried out on superpixel level, the labeled pixels are distributed in different superpixels and the labels of training samples should first spread out within the superpixel scale. Considering the training set of hyperspectral image $\mathbf{R}$, $\mathbf{A}_c = [\mathbf{a}_{c,1}, \mathbf{a}_{c,2}, \ldots, \mathbf{a}_{c,n_c}] \in \mathbb{R}^{B\times n_c}$ is used to represent the training samples from the cth ($c = 1, \ldots, C$) class, where C is the number of surface objects in the scene. Thus, the whole training set $\mathbf{A} \in \mathbb{R}^{B\times n}$ is the concatenation of all training samples, i.e., $[\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_C]$, and $n = n_1 + n_2 + \cdots + n_C$.

In order to keep sample regularization within each superpixel consistent, the circumstance that the superpixel contains labeled samples from the same class is specifically considered, which can be formalized as follows.

1) *C1:* $\mathbf{R}_{(i,j)} \in \mathbf{A}_c, \mathbf{S}_{(i,j)} = k, c = 1, \ldots, C, k = 1, \ldots, K$.
2) *C2:* $\forall \mathbf{a} \in \mathbf{A} - \mathbf{A}_c, \mathbf{a}$ is not inside $\mathbf{S}_k$.

More precisely, condition 1 (C1) claims that the kth superpixel $\mathbf{S}_k$ involves labeled samples (maybe more than one) from the cth class, while condition 2 (C2) indicates that $\mathbf{S}_k$ does not have labeled sample from other class. In brief, when a superpixel contains only labeled samples from a single class, the sample regularization procedure can be applied, which means that the label of all pixels inside the superpixel is assigned to the class of the included labeled sample. Alternatively, for the other circumstances that cannot meet the above two conditions at the same time, we introduced a simple way to further divide the superpixel that contains labeled samples from different classes into separated superpixels. Concerning the situation that there are two training samples from different classes in the same superpixel, the spatial positions of the

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(a) (b)

Fig. 3. Sketch map of superpixel-level sample regularization. (a) Superpixel map before sample regularization. Different color corresponds to different superpixel, while the lattice with a certain number stands for a labeled sample belonging to relevant class. (b) Superpixel map after sample regularization. The label of superpixels that only contains labeled samples from a single class can be determined, while others are resolved by a spatial position-based segmentation scheme.

labeled samples are first investigated. Second, for the unlabeled pixels within the superpixel, they are classified according to the Manhattan distance to the two labeled samples, and the superpixel can be naturally divided into two superpixels. For the other circumstances (more training samples and more classes in the superpixel), the superpixel can be managed in a similar way.

In fact, in our experiments, the circumstance that training samples from different classes in the same superpixel seldom occurs, which is mainly due to the following two reasons. First, we use an over-segmented strategy for superpixel extraction (the scenario of segmenting different types of materials into the same superpixel happens with a low frequency), and the ERS-based superpixel map is generally uniformly distributed (which is advantageous than simple linear iterative clustering [37] and simple noniterative clustering [38]), which means that the sizes of superpixel are comparable with each other. Second, since the small sample set problem is mainly concerned in this article (i.e., the number of labeled samples of each class is scarce), there are very few cases that different classes of training samples are located in the same superpixel.

To make the description more clear, Fig. 3 gives a schematic of the superpixel-level sample regularization procedure. In particular, Fig. 3(a) shows the superpixel map before sample regularization. Here each lattice is a pixel, and the different color corresponds to superpixel coverage. Meanwhile, the lattice with a certain number illustrates a labeled sample. For superpixels that satisfy the two conditions (i.e., C1 and C2), such as the ones with blue, purple, and yellow colors, the sample regularization procedure can be directly applied. Alternatively, for the superpixel with green color, which contains labeled samples from two (classes 1 and 3) different classes, the superpixel is further divided into two new superpixels (brown and green colors). Specifically, the labeled samples are used as the cluster centers, and segmentation is carried out through a Manhattan distance-based reconstruction. After accomplishing sample regularization on the superpixel level, the labels have been spread out in the superpixel scale, as displayed in Fig. 3(b).

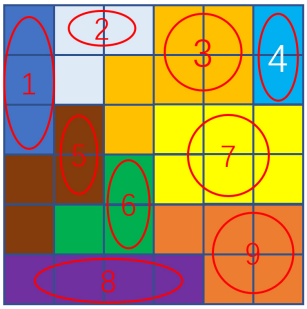


Fig. 4. Sketch map of superpixel neighborhoods. Here, the superpixel map utilized in Fig. 3 is analyzed. The number in each superpixel is the serial number. It can be easily found that the neighborhoods of superpixel $\mathbf{S}_7$ (yellow color) are $\mathbf{S}_3$, $\mathbf{S}_4$, $\mathbf{S}_6$, and $\mathbf{S}_9$, thus $\mathcal{N}_{(7)} = \{\mathbf{S}_3, \mathbf{S}_4, \mathbf{S}_6, \mathbf{S}_9\}$.

Additionally, the label probability matrix $\mathbf{Y}$ defined in Section II-A should be adapted to superpixel-oriented processing (denoted as $\mathcal{Y} \in \mathbb{R}^{K\times C}$). Similarly, $\mathcal{Y}$ is divided into two parts, i.e., labeled superpixels and unlabeled ones, which are, respectively, expressed as $\mathcal{Y}^{(l)}$ and $\mathcal{Y}^{(u)}$. Likewise, $\mathcal{Y}^{(l)}$ is initialized as follows:

$$\mathcal{Y}^{(l)}_{(k,c)} = \begin{cases} 1, & \mathbf{S}_k \text{ is expanded to class } c, \\ & c = 1, \ldots, C, \ k = 1, \ldots, L \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

where L is the number of labeled superpixels through sample regularization. Clearly, $L \leq n$ (recall n is the number of training samples).

C. Superpixel-Based Transition Matrix Construction

Having finished superpixel-oriented feature extraction (recall $\mathbf{G} \in \mathbb{R}^{K\times 2B}$ is the achieved feature matrix) and superpixel-level sample regularization, the superpixel-based similarity matrix $\mathcal{M} \in \mathbb{R}^{K\times K}$ and the corresponding transition matrix $\mathcal{O} \in \mathbb{R}^{K\times K}$ are constructed as follows.

1) Calculation of Superpixel Neighborhoods: Unlike the raw hyperspectral image that the neighborhoods of each pixel are regular (such as four neighboring and eight neighboring), the spatial distribution of superpixel in superpixel map $\mathbf{S}$ is mostly irregular, as shown in Fig. 4, and thus the calculation of superpixel neighborhoods should be peculiarly exploited. First, after the superpixel map $\mathbf{S}$ is extracted by the ERS method from the raw hyperspectral image $\mathbf{R}$, the superpixel neighborhoods are computed one by one. Second, for any superpixel $\mathbf{S}_k, k = 1, \ldots, K$, the corresponding spatial positions of pixels inside $\mathbf{S}_k$ are figured out by a masking operation. Third, by applying a morphological operator (called dilation [39]) on the masked image by a predefined structure element, the superpixel outer boundary is determined by investigating the difference between the dilated image and masked one. Since only one pixel outward the superpixel boundary is enough, a 3×3 matrix with each value equaling 1 is just the predefined structure element. At last, those superpixels that contain pixels in the outer boundary set are identified as the neighborhoods of $\mathbf{S}_k$, which are preserved in $\mathcal{N}_{(k)}$ (here $\mathcal{N}$ denotes the superpixel neighborhood set). Fig. 4 illustrates the distribution of superpixels. It can be observed that the neighborhoods of superpixel $\mathbf{S}_7$ (yellow color) consist

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Algorithm 1 Calculation of Superpixel Neighborhoods

INPUT: raw hyperspectral image $\mathbf{R} \in \mathbb{R}^{X\times Y\times B}$, structure element $\boldsymbol{SE}$ (which is a 3×3 matrix and all elements are set as 1);
OUTPUT: superpixel map $\mathbf{S} \in \mathbb{R}^{X\times Y}$ and superpixel neighborhood set $\boldsymbol{\mathcal{N}}$;
Begin
use equation (9) to compute the number of extracted superpixels K;
apply ERS method on $\mathbf{R}$ to obtain the superpixel map $\mathbf{S} \in \mathbb{R}^{X\times Y}$ with K superpixels;
for $k = 1$ to K **do**
  $\boldsymbol{mask} = \text{zeros}(X, Y)$;
  for $i = 1$ to X **do**
    for $j = 1$ to Y **do**
      if $\mathbf{S}_{(i,j)} == k$
        $\boldsymbol{mask}_{(i,j)} = 1$;
      else
        $\boldsymbol{mask}_{(i,j)} = 0$;
      end
    end for
  end for
  $\phi(\mathbf{S}_k) = \text{imdilate}(\boldsymbol{mask}, \boldsymbol{SE})$ *% the dilation operation in image processing field*;
  outerBoundary $= \phi(\mathbf{S}_k) - \boldsymbol{mask}$;
  find all positions in outerBoundary where the element equal to 1, and the superpixels containing these positions are the neighbors of superpixel $\mathbf{S}_k$, which are preserved in $\boldsymbol{\mathcal{N}}_{(k)}$;
end for

of four superpixels, i.e., $\mathbf{S}_3$, $\mathbf{S}_4$, $\mathbf{S}_6$, and $\mathbf{S}_9$, so $\boldsymbol{\mathcal{N}}_{(7)} = \{\mathbf{S}_3, \mathbf{S}_4, \mathbf{S}_6, \mathbf{S}_9\}$. Besides, the detailed description of superpixel neighborhood extraction is summarized in Algorithm 1.

2) Similarity Matrix Computation Based on Superpixels: Since superpixel reflects the spatial distribution of local region, the similarity measure between superpixels should be specially designed. In this respect, different from the pixel-oriented distance metrics such as Euclidean and Chebyshev distance, the similarity representation between superpixels should be estimated from the whole superpixel viewpoint rather than a single pixel. That is to say, two neighboring superpixels that are highly correlated should have large similarity value as usual, but those having small feature discrepancy should output relatively lower similarity value, and thus the reliability of superpixel-oriented distance measure can be ensured. Based on the intuitive analysis, our superpixel-oriented distance function between any two superpixel feature vector $\mathbf{G}_i \in \mathbb{R}^{2B}$ and $\mathbf{G}_k \in \mathbb{R}^{2B}, i, k = 1, \ldots, K$, is derived as follows:

$$\boldsymbol{\mathcal{M}}(i,k) = \begin{cases} \dfrac{1}{2B} \exp^{-\sum_{b=1}^{2B} \frac{2|\mathbf{G}_i^b - \mathbf{G}_k^b|}{|\mathbf{G}_i^b - \mathbf{G}_k^b| + |\mathbf{G}_i^b + \mathbf{G}_k^b|}}, & \text{if } \mathbf{S}_k \in \boldsymbol{\mathcal{N}}_{(i)} \\ i,\ k = 1, \ldots, K & \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

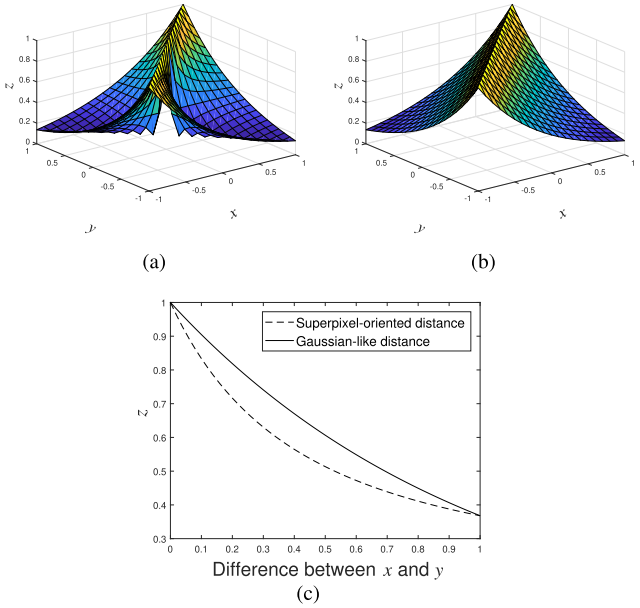


Fig. 5. Illustrations of distance functions. (a) Superpixel-oriented distance function $z = \exp^{-(2|x-y|)/(|x-y|+|x+y|)}$. (b) Gaussian-like distance function $z = \exp^{-\|x-y\|_2^2}$. (c) Comparison of two different distance measures when $x + y = 1$. It can be readily observed that the curve obtained by our superpixel-oriented distance function decreases much faster than the Gaussian-like one.

Obviously, only superpixel similarity values in the neighborhood are reserved, and others are omitted. That is, the strategy constrains the propagation between adjacent superpixels, guaranteeing the effectiveness of the subsequent LP procedure. In particular, in order to make the introduced superpixel-oriented distance function [see (19)] easily to be understood, Fig. 5 illustrates two distance functions (i.e., our proposed one and a Gaussian-like one) in 3-D view. It is worth pointing out that the drawing function displayed in Fig. 5(a) is slightly different from (19), which are, respectively, applied on scalars and vectors. Clearly, the similarity values of our superpixel-oriented distance function [see Fig. 5(a)] decrease much faster than those of the Gaussian-like one [see Fig. 5(b)]. Furthermore, Fig. 5(c) compares the two distance functions when $x + y = 1$. It can be readily observed from Fig. 5(c) that the similarity (i.e., the z-values) achieved by our superpixel-oriented distance measure decreases much faster than the Gaussian-like one as the increase of difference between x and y, which accords with the aforementioned analysis of superpixel distance.

Similarly, (19) can be normalized and the corresponding transition matrix $\boldsymbol{\mathcal{O}} \in \mathbb{R}^{K\times K}$ is computed as follows:

$$\boldsymbol{\mathcal{O}}(i,k) = \frac{\boldsymbol{\mathcal{M}}_{(i,k)}}{\sum_{\mathbf{S}_k \in \boldsymbol{\mathcal{N}}_{(i)}} \boldsymbol{\mathcal{M}}_{(i,k)}}, \quad i,\ k = 1, \ldots, K. \quad (20)$$

D. Weighted LP

It can be easily found from the DLP method presented in Section II-B that two parameters (λ and β) should be predefined to regularize label correlation and identity matrix,

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respectively. Since both parameters are artificially specified and cannot readily adapt to hyperspectral images with various spatial coverage and resolution, the generalization ability of DLP is deteriorated. Alternatively, an SuWLP approach is proposed to automatically decide the two parameters by taking advantage of superpixel properties. More precisely, except the mean vector $\mathbf{M}_k \in \mathbb{R}^B, k = 1, \ldots, K$ extracted from each superpixel, the standard deviation is also an important statistical description that reflects the degree of dispersion within superpixel. Since the spatial distribution of superpixel in every spectral band is similar, the mean standard deviation σ_k of superpixel $\mathbf{S}_k$ can be thus defined as

$$\sigma_k = \frac{1}{B}\sum_{b=1}^{B}\sqrt{\frac{1}{m_k}\sum_{i,j}\left(\mathbf{R}_{(i,j,b)} - \mathbf{M}_k^b\right)^2} \quad \mathbf{S}_{(i,j)} = k \quad k = 1, \ldots, K. \tag{21}$$

Furthermore, the mean standard deviation vector $\boldsymbol{\sigma} \in \mathbb{R}^K$ for the whole superpixel map $\mathbf{S}$ is

$$\boldsymbol{\sigma} = [\sigma_1, \sigma_2, \ldots, \sigma_K]^T. \tag{22}$$

Obviously, a small value of standard deviation $\sigma_k, k = 1, \ldots, K$ represents high homogeneity of superpixel structure. On the contrary, the larger the standard deviation, the stronger the heterogeneity within superpixels. Concerning LP, superpixel with low mean standard deviation should clearly be given a large weight during the propagation process. Therefore, we design a weight function to acquire the weight parameter $\boldsymbol{\omega} \in \mathbb{R}^K$, and the definition is

$$\boldsymbol{\omega} = \exp(-\boldsymbol{\sigma}). \tag{23}$$

With respect to the exponential function $\exp(-\boldsymbol{\sigma})$ (which is a monotonically decreasing function), the weight of the superpixel decreases as the standard deviation increases (corresponding to the high homogeneity of superpixel structure). Besides, it is worth pointing out that the range of (23) is between 0 and 1 since $\sigma_k, k = 1, \ldots, K$ is always greater than (or equal to) 0. On the whole, by incorporating the standard deviation of superpixels, the weight parameters can be automatically calculated and easily generalized for different hyperspectral image data sets without any manual intervention.

After all the superpixel-based information has been extracted, including the label probability matrix $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{K\times C}$, the transition matrix $\boldsymbol{\mathcal{O}} \in \mathbb{R}^{K\times K}$, and the weight parameter $\boldsymbol{\omega} \in \mathbb{R}^K$, the iteration process of LP is defined as follows:

$$\boldsymbol{\mathcal{O}}_t = \boldsymbol{\mathcal{O}}\left(\boldsymbol{\mathcal{O}}_{t-1} + \mathrm{diag}(\boldsymbol{\omega})\,\boldsymbol{\mathcal{Y}}_{t-1}\boldsymbol{\mathcal{Y}}_{t-1}^T\right)\boldsymbol{\mathcal{O}}^T + \mathrm{diag}(\boldsymbol{\omega})\mathbf{I} \tag{24}$$

where $\mathrm{diag}(\cdot)$ means the diagonal matrix of $\boldsymbol{\omega}$. Likewise, the iteration process is executed for T times until a stable condition has been reached. Instead of specifying a fixed iteration number for all data sets, a criterion called change rate is introduced to determine whether the iteration procedure should stop. Assuming that the classification map acquired from label probability matrix $\boldsymbol{\mathcal{Y}}_t$ and $\boldsymbol{\mathcal{Y}}_{t-1}$ are $\mathcal{L}_t \in \mathbb{R}^{X\times Y}$ and $\mathcal{L}_{t-1} \in \mathbb{R}^{X\times Y}$, respectively, the change rate can be computed as

$$C(\boldsymbol{\mathcal{Y}}_t, \boldsymbol{\mathcal{Y}}_{t-1}) = \frac{\sum_{i=1}^{X}\sum_{j=1}^{Y} I(\mathcal{L}_t(i,j), \mathcal{L}_{t-1}(i,j))}{X \times Y} \tag{25}$$

where I is the indicator function and defined as

$$I(t_1, t_2) = \begin{cases} 1, & \text{if } t_1 \neq t_2 \\ 0, & \text{otherwise.} \end{cases} \tag{26}$$

Meanwhile, $\mathcal{L}_t(i,j)$ and $\mathcal{L}_{t-1}(i,j)$ are, respectively, the class label of the tth and $(t-1)$th iteration at the coordinates (i,j). Concretely, $\mathcal{L}_t(i,j)$ is determined by

$$\mathcal{L}_t(i,j) = \{\arg\max_{c=1,\ldots,C} \boldsymbol{\mathcal{Y}}_t(k,c) | \mathbf{S}_{(i,j)} = k\}. \tag{27}$$

That is, the predict label of $\mathcal{L}_t(i,j)$ is just the label of superpixel $\mathbf{S}_k$ that pixel (i,j) belongs to, which is the class corresponding to the column where the maximal value of the kth row in label probability matrix $\boldsymbol{\mathcal{Y}}_t$.

In the experiments, a small threshold ϵ is predefined (usually 10^{-2}) and the LP procedure stops when $C(\boldsymbol{\mathcal{Y}}_t, \boldsymbol{\mathcal{Y}}_{t-1})$ is smaller than ϵ, and the final classification map $\mathcal{L}$ can be thus achieved. The pseudocode of the proposed SuWLP algorithm is outlined in Algorithm 2.

Algorithm 2 SuWLP for Hyperspectral Image Classification

INPUT: raw hyperspectral image $\mathbf{R} \in \mathbb{R}^{X\times Y\times B}$, training set $\mathbf{A} \in \mathbb{R}^{B\times n}$ from C classes, $\epsilon = 10^{-2}$;
OUTPUT: classification map $\mathcal{L}$;
Begin
utilize Algorithm 1 to acquire the superpixel map $\mathbf{S} \in \mathbb{R}^{X\times Y}$ (containing K superpixels) and superpixel neighborhood set $\mathcal{N}$;
use equation (17) to create superpixel feature $\mathbf{G} \in \mathbb{R}^{K\times 2B}$;
initialize label probability matrix $\boldsymbol{\mathcal{Y}}_0 = [\boldsymbol{\mathcal{Y}}_0^{(l)}; 0] \in \mathbb{R}^{K\times C}$, where $\boldsymbol{\mathcal{Y}}_0^{(l)}$ is set by equation (18);
construct the superpixel similarity matrix $\boldsymbol{\mathcal{M}}$ and initial transition matrix $\boldsymbol{\mathcal{O}}_0$ by equation (19) and (20), respectively;
calculate the weight parameter $\boldsymbol{\omega} \in \mathbb{R}^K$ by equation (23);
$t = 1$;
do
  $\boldsymbol{\mathcal{Y}}_t = \boldsymbol{\mathcal{O}}_{t-1} * \boldsymbol{\mathcal{Y}}_{t-1}$;
  $\boldsymbol{\mathcal{Y}}_{(t)}^{(l)} = \boldsymbol{\mathcal{Y}}_0^{(l)}$;
  $\boldsymbol{\mathcal{O}}_t = \boldsymbol{\mathcal{O}}(\boldsymbol{\mathcal{O}}_{t-1} + diag(\boldsymbol{\omega})\boldsymbol{\mathcal{Y}}_{t-1}\boldsymbol{\mathcal{Y}}_{t-1}^T)\boldsymbol{\mathcal{O}}^T + diag(\boldsymbol{\omega})\mathbf{I}$;
  $t = t + 1$;
while$(C(\boldsymbol{\mathcal{Y}}_t, \boldsymbol{\mathcal{Y}}_{t-1}) > \epsilon)$
use equation (27) to obtain the final classification map $\mathcal{L}$;
End

IV. Experimental Setup

In this section, three commonly used hyperspectral images, Indian Pines, Salinas, and Kennedy Space Center (KSC) are first introduced. Then, the methods for comparison are carefully designed.

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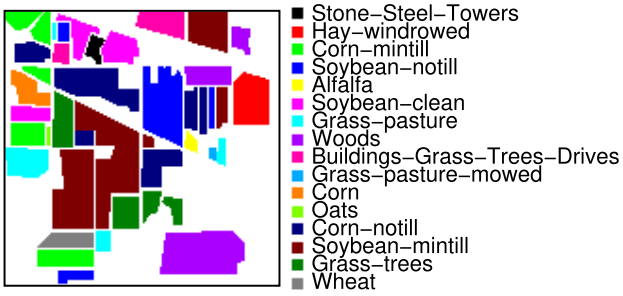


Fig. 6. Groundtruth map of the Indian Pines hyperspectral image (16 land-cover classes).

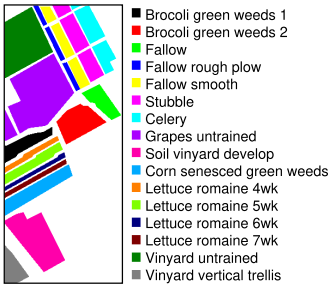


Fig. 7. Groundtruth map of the Salinas hyperspectral image (16 land cover classes).

TABLE I

LAND COVER CLASSES WITH NUMBER OF SAMPLES FOR THE INDIAN PINES HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
$C1$	Stone-steel-towers	95
$C2$	Hay-windrowed	489
$C3$	Corn-min Till	834
$C4$	Soybean-no Till	968
$C5$	Alfalfa	54
$C6$	Soybean-clean Till	614
$C7$	Grass/Pasture	497
$C8$	Woods	1,294
$C9$	Bldg-Grass-Tree-Drives	380
$C10$	Grass/Pasture-mowed	26
$C11$	Corn	234
$C12$	Oats	20
$C13$	Corn-no Till	1,434
$C14$	Soybean-min Till	2,468
$C15$	Grass/Trees	747
$C16$	Wheat	212
	Total	10,366

TABLE II

LAND COVER CLASSES WITH NUMBER OF SAMPLES FOR THE SALINAS HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
$C1$	Brocoli green weeds 1	2,009
$C2$	Brocoli green weeds 2	3,726
$C3$	Fallow	1,976
$C4$	Fallow rough plow	1,394
$C5$	Fallow smooth	2,678
$C6$	Stubble	3,959
$C7$	Celery	3,579
$C8$	Grapes untrained	11,271
$C9$	Soil vinyard develop	6,203
$C10$	Corn senesced green weeds	3,278
$C11$	Lettuce romaine 4wk	1,068
$C12$	Lettuce romaine 5wk	1,927
$C13$	Lettuce romaine 6wk	916
$C14$	Lettuce romaine 7wk	1,070
$C15$	Vinyard untrained	7,268
$C16$	Vinyard vertical trellis	1,807
	Total	54,129

A. Hyperspectral Image Description

The first data set used in our experiment is the Indian Pines image, which was gathered by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the Indian Pines test site in North-western Indian. The image consists of 145 × 145 pixels and the spatial resolution is 20 m per pixel. The wavelength of the AVIRIS sensor ranges from 0.4 to 2.5 μm, which is divided into 224 bands. Before experiments, four zero bands and 35 low signal-to-noise ratio (SNR) bands were removed, and only 185 channels are preserved. There are 16 classes of 10 366 labeled pixels in total. Fig. 6 shows the groundtruth of the Indian Pines hyperspectral image. The distribution of labeled samples for different classes is extremely unbalanced, that is, land covers such as Alfalfa, Grass/Pasture-mowed, and Oats contains only 54, 26, and 20 labeled samples, respectively, which is a big challenge for hyperspectral image classification. Table I displays the detail information of different land cover and the corresponding number of labeled samples.

The second data set Salinas was also acquired by the AVIRIS sensor over Salinas Valley, CA, USA. The area covered comprises 512 lines by 217 samples. Although the data are collected using the same sensors as Indian Pines, Salinas has a higher spatial resolution of 3.7 m per pixel.

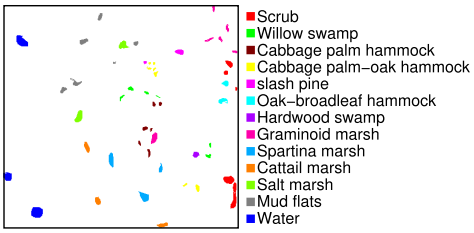


Fig. 8. Groundtruth map of the KSC hyperspectral image (13 land cover classes).

Same as the Indian Pines data set, 20 water absorption bands are prediscarded, and the retained 204 bands were used for experiments. This scene covers soils, vegetables, and vineyard fields. The reference land cover map is illustrated in Fig. 7. Among all the 111 104 pixels, nearly half of the pixels (54 129) are labeled, and the detail information for the labeled samples is provided in Table II.

The third data set KSC was collected by the AVIRIS sensor over the KSC, FL, USA, on March 23, 1996. The KSC data contain 221 bands and with a spatial size of 421 × 444. The spatial resolution is close to the Indian Pines data set, i.e., 18 m per pixel. After removing water absorption and low SNR bands, 176 bands were kept for the subsequent processing.

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TABLE III
LAND COVER CLASSES WITH NUMBER OF SAMPLES FOR THE KSC HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of sample
C1	Scrub	761
C2	Willow swamp	243
C3	Cabbage palm hammock	256
C4	Cabbage palm/oak hammock	252
C5	Slash pine	161
C6	Oak/broadleaf hammock	229
C7	Hardwood swamp	105
C8	Graminoid marsh	431
C9	Spartina marsh	520
C10	Cattail marsh	404
C11	Salt marsh	419
C12	Mud flats	503
C13	Water	927
	Total	5211

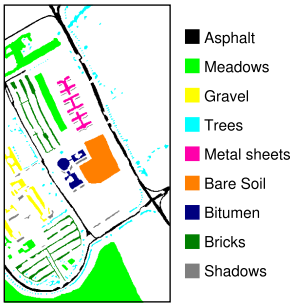


Fig. 9. Groundtruth map of the Pavia University hyperspectral image (nine land cover classes).

There are very few labeled samples in the KSC data set, and 5122 labeled samples are scattered among 13 different classes. Fig. 8 shows the groundtruth and the number of labeled sample for different classes are listed in Table III.

The last data set is an urban cover called Pavia University image, which was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) airborne hyperspectral sensor. The image contains 610×340 pixels with a spatial resolution of 1.3 m. This image consists of 115 bands originally, and 13 spectral bands are removed due to noise. For classification, it comprises nine land cover classes and 42 776 labeled samples. Likewise, the groundtruth map and detailed information about the labeled samples are, respectively, presented in Fig. 9 and Table IV.

B. Experimental Design

In order to verify the validity of the proposed semi-supervised classification approach called SuWLP, several existing methods such as support vector machine with raw hyperspectral (SVM-RAW) and extended multiattribute profile (SVM-EMAP), spatial–spectral-integrated techniques including generalized composite kernel (GCK) machines [40] and local binary pattern-based extreme learning machine (LBP-ELM) [41], are utilized in comparison. Besides, the superpixel version of DLP [24], named as SuDLP, is also considered in the experiments because pixel-level DLP is infeasible in hyperspectral image classification.

TABLE IV
LAND COVER CLASSES WITH NUMBER OF SAMPLES FOR THE PAVIA UNIVERSITY HYPERSPECTRAL IMAGE

Class	Land Cover Type	No. of Samples
C1	Asphalt	6,631
C2	Meadows	18,649
C3	Gravel	2,099
C4	Trees	3,064
C5	Painted metal sheets	1,345
C6	Bare Siol	5,029
C7	Bitumen	1,330
C8	Self-Blocking Bricks	3,682
C9	Shadows	947
	Total	42,776

Since this article is mainly dedicated to exploiting small sample problems, only 3–15 labeled samples from each class are randomly selected to constitute the training set and the rest of labeled samples are used for testing. Meanwhile, each circumstance is executed ten times, while the mean values and standard deviation are reported. Besides, overall accuracy (OA) and the Kappa coefficient (κ) are used to evaluate the classification performance obtained by various methods. The detailed definition of OA and κ can be found in [42].

Before entering Section V, the impact of the iteration procedure of LP is analyzed. As listed in Section I, one main advantage of our proposed SuWLP approach is the method robustness, which means that all the parameters contained in SuWLP can be preset and adapted to hyperspectral images with various spectral and spatial resolutions. Meanwhile, it can be found from Algorithm 2 that the iterative process is the most time-consuming part. To address this issue in more detail, Fig. 10 illustrates the impact of iteration number on the change rate defined in (25). Here, the change rate is the mean values of the ten experiments on each data set (here three samples per class as the training set is concerned). As we can see, the change rate on the four data sets generally decreases as the increase of the iteration number (it is worth pointing out that the very high change rate, nearly 100%, at initial iterations corresponds to LP procedure from labeled samples to unlabeled ones). Also, the change rate of Indian Pines, Salinas, and Pavia University hyperspectral images drops much faster than that of the KSC data set. In particular, since the termination condition mainly investigates the classification difference between two successive iterations, which should be less than a predefined threshold (such as 10^{-2} used in Algorithm 2), the number of iterations does not need to be specified. Obviously, as displayed in Fig. 10, ten iterations of the proposed weighted LP are enough for the Indian Pines, Salinas, and Pavia University data set, whereas the KSC data set needs about 20 iterations. Moreover, because the weighted LP is carried out on the superpixel level, the proposed SuWLP approach is thus applicable for the hyperspectral image with large spatial coverage.

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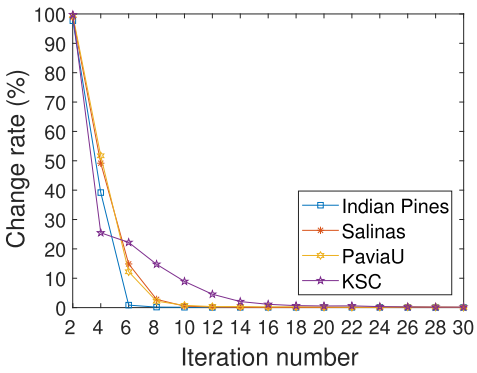
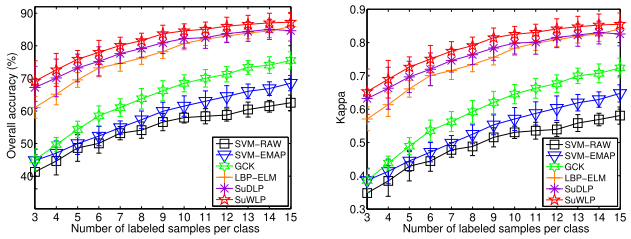


Fig. 10. Impact of iteration number on the change rate (three samples per class as the training set).



(a) (b)

Fig. 11. Indian Pines hyperspectral image. (a) OA and (b) Kappa as functions of the number of labeled samples per class.

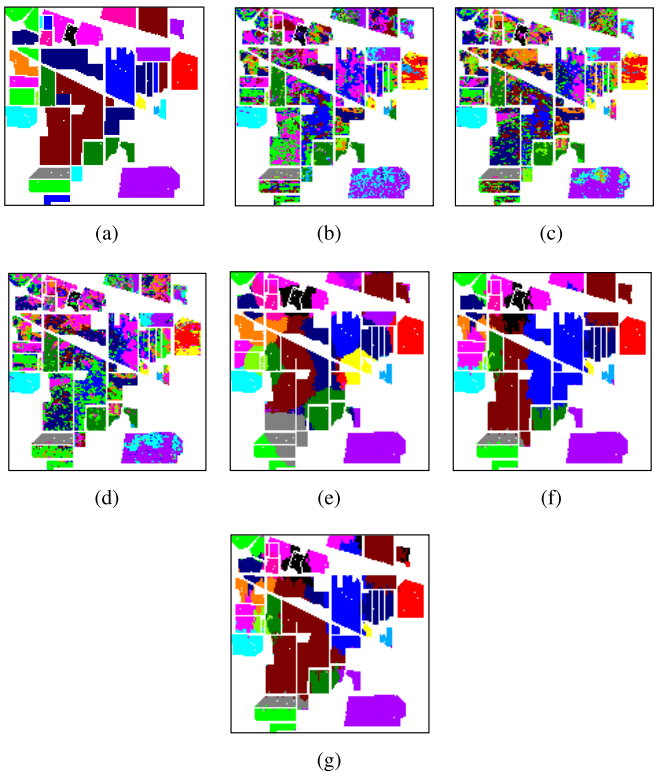
V. Experimental Results

In this section, experimental results on Indian Pines, Salinas, and KSC data sets with various methods have been discussed, and the superiority of the proposed SuWLP approach has been demonstrated.

A. Indian Pines Hyperspectral Image

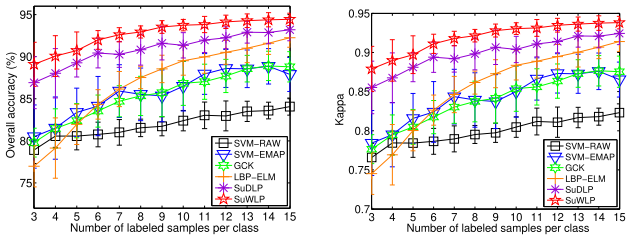
The classification accuracy of SuWLP and other compared methods, including SVM-RAW, SVM-EMAP, GCK, LBP-ELM, and SuDLP, with a different number of training samples per class is shown in Fig. 11, in which OA and Kappa measures are both presented. As usual, the training size has a positive impact on the classification performance of all the compared methods. More precisely, the accuracies obtained by SVM-EMAP are better than SVM-RAW, showing the necessity of feature extraction. Meanwhile, the two superpixel-oriented methods, both SuDLP and SuWLP, achieve better performance than GCK and LBP-ELM, indicating the effectiveness of graph-based semisupervised learning methods. Moreover, our proposed SuWLP approach also performs better than SuDLP, validating the efficiency of the adaptive superpixel-related parameter estimation scheme.

Furthermore, in order to exhibit the small sample size problem more clear, Table V lists the classification performance of compared methods, including mean accuracy and standard deviation of all the 16 land covers, OA and Kappa, when only three labeled samples per class constitute the training set. It can be readily found from Table V that the classification accuracies achieved by our SuWLP approach are higher than



(a) (b) (c) (d) (e) (f) (g)

Fig. 12. Indian Pines hyperspectral image (three labeled samples per class for training). (a) Testing set. (b) SVM-RAW (38.86%). (c) SVM-EMAP (40.30%). (d) GCK (39.32%). (e) ELM-LBP (61.02%). (f) SuDLP (64.85%). (g) SuWLP (71.18%).



(a) (b)

Fig. 13. Salinas hyperspectral image. (a) OA and (b) Kappa as functions of the number of labeled samples per class.

other methods of comparison in most circumstances. Furthermore, the classification maps of test samples obtained by various methods in a single experiment are visually provided in Fig. 12. Likewise, the classification map obtained by SuWLP [see Fig. 12(g)] is in more accordance with the test set [see Fig. 12(a)].

B. Salinas Hyperspectral Image

Same as the Indian Pines hyperspectral image, a fixed number of labeled samples, i.e., 3–15 samples per class, is picked out to form the training set and the rest ones are used to evaluate the various compared techniques, as displayed in Fig. 13. From Fig. 13, it can be clearly seen that when the training sample is only three per class, the two superpixel-oriented algorithms, both SuDLP and SuWLP, have

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TABLE V

CLASSIFICATION PERFORMANCE USING THE SIX COMPARED METHODS ON THE INDIAN PINES HYPERSPECTRAL IMAGE WITH THREE LABELED SAMPLES PER CLASS AS TRAINING SET (NUMBERS IN BOLD REPRESENT THE BEST CLASSIFICATION PERFORMANCE)

Class	SVM-RAW		SVM-EMAP		GCK		LBP-ELM		SuDLP		SuWLP	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	82.72	13.95	92.72	5.13	74.89	6.84	95.43	6.83	90.87	15.50	**97.28**	4.29
C2	47.51	13.50	57.53	16.31	52.18	10.77	97.55	2.64	97.49	2.54	**98.56**	1.59
C3	29.86	13.28	26.76	12.21	31.66	14.67	44.13	12.50	50.84	13.59	**53.73**	16.36
C4	43.60	15.77	38.56	14.03	35.58	14.37	61.53	7.93	70.37	17.35	**73.41**	13.35
C5	60.59	14.18	63.73	12.85	58.82	12.33	**99.41**	1.86	76.08	11.50	92.94	9.34
C6	25.43	10.41	25.29	6.99	27.89	12.23	53.24	11.33	**58.66**	16.18	55.71	17.12
C7	59.88	15.14	69.64	8.14	66.36	6.04	61.90	7.74	**74.51**	10.74	73.52	10.52
C8	60.68	16.43	69.37	13.55	67.17	8.02	**88.13**	8.49	83.35	10.06	86.26	11.82
C9	22.94	9.39	34.27	11.77	28.67	13.17	77.14	22.86	81.11	23.41	**81.72**	23.30
C10	86.52	13.97	90.00	6.50	91.74	4.32	**99.57**	1.37	86.52	17.20	93.48	11.27
C11	37.53	8.57	35.97	8.79	38.31	12.47	**88.23**	11.95	49.22	10.54	69.31	11.60
C12	92.94	12.34	82.94	15.05	83.53	12.34	**97.06**	6.35	85.29	11.52	92.94	11.37
C13	19.93	9.79	24.40	6.75	28.69	11.05	**39.16**	12.14	34.46	11.98	38.85	9.47
C14	35.05	13.00	37.13	10.05	38.99	14.53	48.02	8.55	73.74	21.52	**76.54**	18.53
C15	63.60	16.87	**72.63**	8.65	70.66	8.30	67.06	13.04	63.90	17.97	54.13	13.91
C16	91.15	4.48	89.14	8.30	89.62	9.57	88.76	12.39	93.83	2.72	**94.78**	2.07
OA	41.32	5.23	44.82	3.40	44.98	3.41	61.10	3.38	67.23	6.49	**69.17**	6.23
Kappa	0.35	0.05	0.39	0.04	0.39	0.03	0.57	0.04	0.63	0.07	**0.65**	0.07

TABLE VI

CLASSIFICATION PERFORMANCE BY USING THE SIX COMPARED METHODS FOR THE SALINAS HYPERSPECTRAL IMAGE WITH THREE LABELED SAMPLES PER CLASS AS TRAINING SET (NUMBERS IN BOLD REPRESENT THE BEST CLASSIFICATION PERFORMANCE)

Class	SVM-RAW		SVM-EMAP		GCK		LBP-ELM		SuDLP		SuWLP	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	96.48	4.47	91.55	10.89	92.79	11.65	97.87	2.11	99.52	0.27	**99.67**	0.17
C2	94.25	7.99	98.34	1.47	90.87	5.86	85.41	4.28	99.98	0.05	**100.00**	0.01
C3	73.58	16.92	78.68	20.29	86.28	12.10	84.65	12.24	**99.78**	0.24	**99.78**	0.24
C4	97.44	1.76	98.38	0.97	92.77	5.84	96.26	3.95	95.51	4.89	**98.55**	1.65
C5	91.35	8.14	91.59	6.72	**93.13**	5.47	84.28	11.02	83.87	8.20	86.34	4.83
C6	**97.80**	2.20	96.67	1.93	94.00	3.64	78.48	11.03	95.93	1.77	96.55	0.57
C7	99.13	0.59	99.70	0.11	98.92	0.48	74.06	9.73	99.19	1.26	**99.80**	0.08
C8	55.41	17.21	46.94	15.93	46.47	15.11	72.39	14.66	79.01	10.29	**85.12**	12.30
C9	96.29	2.24	94.40	1.28	99.34	0.26	75.30	12.00	99.42	1.11	**99.60**	0.84
C10	71.78	9.47	78.91	11.37	**85.76**	5.14	79.33	8.70	71.44	9.13	69.71	7.60
C11	84.51	5.91	90.22	5.24	**90.48**	2.38	90.32	6.87	46.75	16.26	47.79	16.02
C12	98.59	1.31	**99.95**	0.11	99.11	1.25	78.07	8.57	62.35	15.39	68.00	16.07
C13	98.01	1.07	98.26	0.99	**99.26**	0.30	66.08	14.31	33.08	8.80	32.39	8.76
C14	88.85	4.28	**91.12**	2.75	82.52	5.60	79.76	5.77	67.32	11.58	73.48	13.86
C15	53.08	24.27	69.55	16.25	65.43	15.75	64.00	11.67	89.10	10.00	**91.85**	5.55
C16	83.38	6.14	96.90	2.17	85.94	6.39	82.09	14.95	**100.00**	0.00	**100.00**	0.00
OA	78.91	1.61	80.51	3.75	79.79	1.72	76.98	2.46	86.91	3.09	**89.09**	2.62
Kappa	0.77	0.02	0.78	0.04	0.78	0.02	0.75	0.03	0.86	0.03	**0.88**	0.03

a large improvement compared to other methods, which is nearly ten percentage. Looking back at the groundtruth of Salinas (see Fig. 7), we can find that the spatial distribution of objects is very regular, and thus the extracted superpixel map can well characterize the spatial structures of land covers, which could greatly increase the efficiency of subsequent LP process. Therefore, it is reasonable to conclude that regular spatial distribution could have a positive effect on classification performance. Similarly, Table VI presents the detailed classification accuracies of the compared methods when only three labeled samples per class are used as the training set, and our proposed SuWLP approach gives the best performance in most cases. Hereafter, the classification maps are omitted due to the similar results.

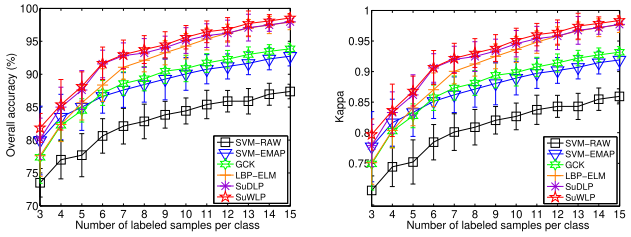


Fig. 14. KSC hyperspectral image. (a) OA and (b) Kappa as functions of the number of labeled samples per class.

C. KSC Hyperspectral Image

Fig. 14 illustrates the classification results of the proposed approach and other five methods in comparison, where the

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TABLE VII

CLASSIFICATION PERFORMANCE BY USING THE SIX COMPARED METHODS FOR THE KSC HYPERSPECTRAL IMAGE WITH THREE LABELED SAMPLES PER CLASS AS TRAINING SET (NUMBERS IN BOLD REPRESENT THE BEST CLASSIFICATION PERFORMANCE)

Class	SVM-RAW		SVM-EMAP		GCK		LBP-ELM		SuDLP		SuWLP	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	79.93	8.46	84.80	6.80	78.69	12.19	69.13	14.69	85.77	12.20	**85.96**	12.37
C2	75.29	10.15	76.50	8.89	**78.67**	12.61	56.21	11.57	73.92	16.60	74.25	16.97
C3	71.58	24.59	87.08	10.53	86.44	7.69	65.73	18.62	84.86	12.00	**87.21**	10.40
C4	40.44	6.51	56.79	17.83	55.82	13.87	53.73	16.56	**76.35**	20.26	75.74	20.55
C5	47.03	16.03	64.81	15.42	55.25	16.56	93.04	8.23	94.24	8.71	**94.96**	8.15
C6	42.35	11.46	64.69	13.15	61.81	19.08	84.69	23.30	96.86	8.45	**97.68**	8.14
C7	89.51	6.40	94.41	7.58	94.41	5.85	**100.00**	0.00	**100.00**	0.00	**100.00**	0.00
C8	46.64	14.85	**67.87**	15.64	61.40	18.71	55.14	13.63	58.95	12.21	61.78	12.40
C9	80.85	14.31	83.09	16.72	**88.67**	8.97	79.52	12.78	71.47	24.85	73.37	23.42
C10	57.66	19.63	55.44	21.14	79.18	11.24	**94.84**	6.41	70.55	9.92	70.65	9.88
C11	87.79	2.87	89.23	3.89	67.57	9.46	82.28	18.74	**91.06**	18.85	**91.06**	18.85
C12	68.22	20.87	**79.22**	8.75	67.76	20.36	64.36	15.32	73.88	13.48	67.40	19.65
C13	99.25	2.00	96.43	11.29	94.92	14.91	**100.00**	0.00	85.80	18.33	95.66	11.23
OA	73.51	2.16	79.92	5.22	77.38	3.95	77.63	3.13	80.20	3.13	**81.84**	2.20
Kappa	0.71	0.02	0.78	0.06	0.75	0.04	0.75	0.04	0.78	0.03	**0.80**	0.03

TABLE VIII

CLASSIFICATION PERFORMANCE BY USING THE SIX COMPARED METHODS FOR THE PAVIA UNIVERSITY HYPERSPECTRAL IMAGE WITH THREE LABELED SAMPLES PER CLASS AS TRAINING SET (NUMBERS IN BOLD REPRESENT THE BEST CLASSIFICATION PERFORMANCE)

Class	SVM-RAW		SVM-EMAP		GCK		LBP-ELM		SuDLP		SuWLP	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	57.93	7.55	53.99	9.37	**58.40**	11.49	46.79	13.87	41.92	11.17	49.56	10.73
C2	51.19	16.68	54.61	17.22	56.99	15.19	45.98	10.83	73.94	6.13	**84.72**	5.48
C3	57.67	16.79	57.05	17.58	63.57	15.05	51.51	10.85	**88.15**	14.83	85.01	11.77
C4	77.03	20.15	**82.74**	18.86	61.69	17.96	45.77	12.01	24.73	11.16	35.03	5.52
C5	94.64	8.83	**99.63**	0.00	51.95	17.24	87.35	16.18	72.00	19.75	86.00	15.63
C6	48.68	16.35	55.58	12.62	44.08	18.73	67.75	16.46	87.32	12.95	**90.78**	10.73
C7	85.40	7.17	88.29	8.18	84.48	13.20	64.90	18.84	79.40	20.62	**89.95**	11.29
C8	57.53	14.41	67.57	11.73	65.33	17.41	**68.44**	13.29	63.23	13.75	64.18	14.67
C9	99.74	0.09	**99.76**	0.19	79.76	11.54	51.27	13.54	16.60	7.90	17.49	8.59
OA	58.15	6.73	61.33	7.33	58.26	6.10	52.86	3.52	65.64	2.13	**70.29**	1.73
Kappa	0.49	0.07	0.53	0.08	0.48	0.06	0.43	0.04	0.56	0.03	**0.61**	0.02

x-axis represents the number of samples per class and the y-axis represents the corresponding OA and Kappa coefficient. Of course, the performance increases with the enlargement of the training set. In the meantime, the results achieved by the two superpixel-oriented methods are better than the compared ones, but the advantages of our SuWLP are not as distinct as those in the above two hyperspectral images. Indeed, it can be observed from the groundtruth map of the KSC data set (see Fig. 8) that 13 surface materials distribute throughout the whole image, and the spatial structure is less regular. Since the superpixel map has a critical impact on the proposed weighted LP strategy, the indistinct spatial structure of land covers could degrade the significance of extracted superpixel map, and eventually affect the capability of the proposed method, which needs further investigation. In a like manner, Table VII reports the OA, Kappa coefficient, and individual classification accuracy for different methods when only three labeled samples per class are used for training, and our SuWLP presents better performance in most cases, demonstrating the superiority of the proposed superpixel-oriented LP strategy.

D. Pavia University Hyperspectral Image

Fig. 15 provides the OA and kappa coefficient curves of the proposed SuWLP and the other five comparison methods.

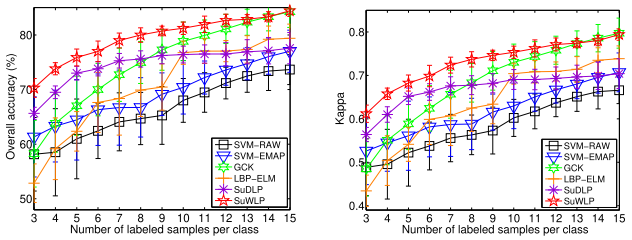


Fig. 15. Pavia University hyperspectral image. (a) OA and (b) Kappa as functions of the number of labeled samples per class.

Similarly, each class randomly selects 3–15 labeled samples as the training set and the rest as test set. Obviously, results obtained by our SuWLP exhibit higher accuracies in most cases. Especially, when the training set is small, the advantages of our proposed method are more obvious. The detailed experimental results of the six methods with only three labeled samples per class for training are listed in Table VIII. Overall, the proposed SuWLP yields better results than other methods. Besides, like the situation encountered in the KSC data set, the spatial distribution of some classes (such as C4 and C9) is scattered, leading to the circumstance that they are indistinct in the superpixel map and eventually deteriorating the classification accuracy, which should be paid more attention.

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VI. Conclusion

In this article, a novel SuWLP approach is proposed for hyperspectral image classification. Instead of conducting the semisupervised learning method, i.e., graph-based LP, on the pixel level, which is infeasible for hyperspectral image processing, a superpixel-based extension is thus presented. In particular, since the superpixel properties (such as shape and neighborhood) are totally different from the pixel viewpoint, a series of superpixel-oriented data processing modules have been subtly designed, including feature computation, neighborhood calculation, and similarity measure. Furthermore, a standard deviation-based parameter estimation scheme is introduced to adaptively determine the weight parameters contained in the LP procedure, and the generalization ability of the proposed SuWLP can be greatly enhanced.

To evaluate the superiority of our SuWLP approach, five state-of-the-art methods in the literature, including SVM-RAW, SVM-EMAP, GCK, ELM-LBP, and the superpixel version of DLP (SuDLP), are compared. Meanwhile, four popular standard hyperspectral image data sets, i.e., Indian Pines, Salinas, KSC, and Pavia University are utilized in the experiments. Numerical and visual experimental results consistently demonstrate the superiority of the proposed SuWLP approach, especially when the training set is small. Besides, since the spatial regularity of land covers can have a critical impact on the classification performance, and the spatial distribution of surface materials is generally closely related, the proposed SuWLP approach can apply for the hyperspectral image in various scenarios. In future work, more attention should be paid to the scattered and irregular materials that may be mis-segmented by the superpixel map.

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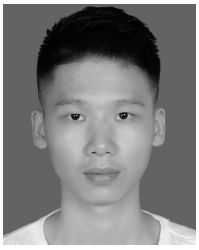
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